

2. [Maximum mark: 5]

Consider the function $f(x) = 2x^4 - 6x^3 + px^2 + qx - 2$, where $p, q \in \mathbb{R}$.
A factor of $f(x)$ is $(x - 1)$, and when $f(x)$ is divided by $(x - 3)$ the remainder is -2 .

Find the value of p and of q .

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